Greyson Ekstrom

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EDUCATION

University of Georgia Expected Graduation: May 2029
B.S. Electrical & Electronics Engineering GPA: 3.84/4.00
Relevant Coursework: Circuit Analysis, Digital Logic Design, Intro to ECSE, Experiential Data Science Specialization

SKILLS

Design & Simulation: Altium Designer, KiCad, LTspice, Multisim, SPICE, PCB Design, Circuit Analysis
Laboratory Equipment: Oscilloscope, Digital Multimeter, DC Power Supply, Soldering
Hardware: Arduino, Raspberry Pi, Jetson Nano, Sensor Integration, Motor Drivers, Encoders
Programming: C++, Python, MATLAB, Java, Digital Logic

PROJECTS

Robotic Arm Dec. 2025 - Present
- Integrated Python OpenCV with YOLOv8 Nano vision processing model on Raspberry Pi which calculates and autonomously manages inverse kinematic motor controls
- Building 4-axis robotic arm controller on Arduino with CNC shield, using AccelStepper to drive stepper motors

Voltage Indicator Sep. 2026 - Present
- Designed a high-voltage detection circuit in Altium Designer for the FSAE EV tractive system, integrating a wide range buck converter and comparator to detect voltages above 60 VDC; completed PCB layout and prepared design for fabrication.
- Simulated circuit performance in LTspice across a 19–400 VDC input range, evaluating 12 V rail stability and threshold detection behavior

Brake System Plausibility Device Sep. 2025 - Aug. 2026
- Designed a Brake System Plausibility Device (BSPD) for the UGA Formula SAE Electric vehicle to trigger shutdown within 450 ms when braking and tractive-system current occur simultaneously, in accordance with FSAE requirements
- Simulated optimized component selection using LTspice and Altium Designer, reducing simulated switching time by 20%

ENGINEERING EXPERIENCE

FSAE EV Powertrain Controls Lead, UGA Motorsports Aug. 2026 - Present
- Led a team of 10 undergraduates to integrate the vehicle's sensor circuitry: selecting and validating multiple components to meet FSAE EV rules and advance development of the University of Georgia's inaugural Formula SAE EV vehicle
- Engineered low voltage sensor processing circuitry with Altium Designer utilizing RC filtering to steady power and reduce noise from analog sensor output voltages resulting in cleaner system readings

FSAE EV Electronics Engineer, UGA Motorsports Sep. 2025 - May 2026
- Collaborated among mechanical and electrical teams to develop a printed circuit board using Altium Designer, fulfilling mechanical integration requirements and competition design regulations
- Developed sensor detection circuit using Altium Designer, simulated various inputs with LTspice ensuring functioning schematic design, optimized component selection reducing net power consumption

IEEE Hardware Competition Electrical Team, UGA Robotics Sep. 2026 - Present
- Co-designed electrical system for an autonomous robot with an 8-person team, utilizing system schematic design, cost vs points analysis, and competition strategy ultimately optimizing design for anticipated 40 second game completion
- Integrated motors, encoders, motor drivers, IMU, servos, TOF sensor network, and IR sensor network; managed I2C and CSI robot serial compatibility creating a 100% electronically compatible sensor network for mechanical and software integration

RoboDawg Electrical Team, UGA Robotics Aug. 2025 - May 2026
- Implemented software and electrical communication between motor controllers and onboard Jetson Nano using Linux
- Set up, tested, and troubleshot persistent software errors and faulty hardware systems during initialization

Software Lead, Flying Legion FIRST Robotics Aug. 2023 - May 2025
- Led a team of 8 developers to design and optimize Java robotic control software and autonomous operation using PhotonVision camera processing, delivering the system within a 6-week deadline and increasing scoring speeds by 60% from season start
- Implemented subsystem control software communicating with brushless motors, encoder feedback loops, PID controls, and servos using Git multibranch workflow to streamline integration and improve subsystem functionality

OTHER EXPERIENCE

Summer Camp Coach, Warner Robins BEST May 2026 - Aug. 2026
- Mentored groups of campers, fostering teamwork, leadership, and positive social development in an inclusive environment
- Maintained camper safety, resolved conflicts, and communicated with staff and campers to support a positive experience

AWARDS

University of Georgia Dean's List Dec. 2025, May 2026